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Task Description:	The overall objective of this NASA Specialized Center of Research (NSCOR) is to model the relative risk of colonic and stomach tumorigenesis for high priority space radiation beams and compare to gamma radiation where human epidemiologic data are available. It will rely primarily on mouse models of gastrointestinal (GI) cancer and human colonic epithelial cell (HCEC) approaches that have been optimized for low dose exposures. This will involve quantitative and qualitative analyses of tumors as well as normal GI epithelium after radiation. Results will be extrapolated to radiation-induced molecular changes in state-of-the-art 2D and 3D cultured human colonocytes. While limited studies at select loci, such as HRPT (hypoxanthine phosphoribosyltransferase), have shown HZE (high energy particle)-induced mutations including splicing errors, frameshifts, and complex changes, we will carry out genome-wide analysis using the power of modern nextgen sequencing (NGS) to assess the spectrum of mutations in radiation-induced tumors, and also persistent genome-wide epigenomic changes in normal cells after ionizing radiation (IR). These NGS omics findings will be integrated with transcriptomics and metabolomics results as well as events at the protein and cellular levels to develop a systems biology model for GI tumorigenesis. By assessing both quantitative in vivo tumorigenesis data and molecular events in mouse tissues and human colonocytes, space radiation-induced tumorigenesis will be compared to human epidemiologic data for low LET (linear energy transfer) radiation exposures, and approaches will be developed to model human risks for GI cancer by space radiation. We continue to employ two models of GI cancers: the APC1638N model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa1638. These mice develop very few adenomas in the small as well as in the large intestine but post-radiation tumor numbers show marked increases. Radiation induced substantial gastric tumorigenesis as well in this model. The second model is the CDX2P Apc flox/+ that has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent a low rate of adenocarcinomas in the colon that is increased post-irradiation. Data being collected from these two relevant mouse models will be used to calculate risk estimates for NASA. Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation compared to the standard risks based on atomic bomb survivors and other exposure populations. In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, in vitro human cell culture and animal models for CRC initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer-related endpoints and systems biology (genomics, proteomics, and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The studies of low- and high-LET radiation on colonocytes gain more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation. Plans for this proposal build on our past accomplishments with GI tumorigenesis models and are focused on studies at low doses with an expanded range of NASA's high priority beams. This proposal is also focused on assessing the effects of gender, dose-rate, age, and genetic heterogeneity in mouse models. Molecular events triggered by space radiation will be investigated in depth and the mechanistic basis for persistent aberrant cell-signaling after HZE radiation will be dissected. Genome-wide NGS combined with other omics approaches will contribute to a systems biology understanding of the effects of space radiation in the intestines and stomach. Both quantitative murine tumor data and results for underlying molecular events in mouse and human GI cells will contribute to modeling of human risks for GI cancer by space travel. The overall aims of this NSCOR are summarized below. Aim 1 (Project 1). Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2 (Project 2). Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3 (Project 3): Characterization of IR-induced neoplastic events in normal diploid human colonocytes. Aim 4 (Project 4). Development of systems biology (Project 4A) and mathematical modeling (Project 4B) approaches for GI cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 132,700 new cases of colorectal cancer (CRC) with 49,700 persons predicted to die of the disease in 2015. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and colorectal carcinogenesis.
	The overall plan is to apply the well-characterized APC mouse models (APC1638N/+ and CDX2P Apcflox/+) of colorectal cancer (CRC) for development of risks estimates for various radiation exposures during manned space flight. We already know from studies in atomic bomb survivors that radiation is risk factor for CRC. Space radiation with its novel characteristics is expected to have greater risk relative to radiation on Earth. A standard approach in radiation biology towards risk estimation is to determine relative biologic effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. However, currently, there is much uncertainty in predicting risk of CRC in astronauts undertaking long duration space missions due to lack of sufficient in vivo mechanistic data in human or in animal models representing the human disease. Considering the limitation of acquiring human data, our study is focused on acquiring sufficient data in mouse models of human CRC with an aim to aid in risk model development for safe human space exploration. Therefore, qualitative and quantitative data on intestinal tumorigenesis in mice generated in this project will be used for quantitative dose response analyses. While such analyses will provide an estimate of risk in mouse model, our ultimate aim is to combine mouse model data from space radiation exposure and human

Task Progress:	epidemiological data from atomic bomb survivors to develop mathematical risk prediction model of CRC for astronauts. Task progress on GI tumorigenesis (GU). We have participated in three beam runs at NASA Space Radiation Laboratory (NSRL) after our current NSCOR award in 2015 with an aim to assess tumorigenesis at doses lower than the previous ones with a broader range of ions. Greater tumorigenesis was observed with 28Si ions in our previous studies relative to other ions and gamma rays. This year we exposed APC1638N/+ mice to 5 cGy of 28-Si (300 MeV/n) and we used male mice due to higher tumor frequency in male relative to female mice. This year we have also started experiments with 16-O (325 MeV/n) and 4-He (250 MeV/n) ions, which are NASA prioritized ions of lower Z value proposed in the NSCOR. In 2017, we are also approved for proton, 28-Si, 56-Fe, and 4-He beam time and we will use the time to irradiated APC1638N/+ mice to fulfill statistical requirements. Importantly, we completed animal numbers required for our low dose rate (0.33 cGy/min) studies, data has been compiled, and tumor data from low and high (50 cGy/min) dose rate exposures were compared. We observed that differences in dose rate did not affect tumor frequency or size. An important aspect of our proposal was to study effects of genetic heterogeneity on space radiation-induced intestinal tumorigenesis. Our backcrossing of APC1638N/+ of C57BL/6J background into C3H/HeN and BALB/c background for the genetic heterogeneity study is near completion. The backcrossing was done for 10 generations and the plan is to sequence APC gene locus for detection of the mutation. We expect APC1638N/+ mice in C3H/HeN and BALB/c background to be ready for space radiation exposure during fall 2017 beam run. We simulated a solar particle event (SPE) at the NSRL to characterize the biological effects of low dose rate protons in vivo. Using the colorectal cancer susceptible (CPC;Apc) mouse model, we studied colonic tumorigenesis after whole-body exposure to a simulated SPE with varying energies (50-150 MeV/n) using a total dose of 2 Gy over a 2 hour period (at an average dose rate of 1.67 cGy/min). We also exposed mice to 2 Gy at 20 cGy/min of acute (50 MeV/n) proton (91% of the sSPE was at 50 MeV/n) or X-ray (250 kVp, 1 mA, 1.65 mm Al filter) at a dose rate of 20 cGy/min as a reference radiation. We observed that whole-body irradiation with simulated SPE was more effective in inducing invasive adenocarcinoma incidence compared to unirradiated controls and x-rays. We started a pilot GCR simulation study in 2016 using protons, and helium and silicon ions. Using the large beam and large boxes to contain 2-3 mice each we were able to irradiate about 90 mice in about 90 minutes with the new fast switching capabilities at the NSRL. We reduced the dose rate to 0.5 cGy per minute (instead of the usual 10-20 cGy per minute) with a total dose of 30 cGy. For NRSL16C we used 20 cGy of proton, followed by 5 cGy of helium and then 5 cGy of silicon. Since 5 cGy of silicon did not increase polyp formation, these experiments were conducted to determine if adding both helium and protons to 5 cGy of silicon would increase polyp formation. We also reversed the order of ions in the GCRsim since it may affect carcinogenesis initiation and/or progression. Results from these experiments will require about 9 additional months. Going forward we will add more mice to this GCRsim for NSRL 17A. This year we focused on assessing long-term effects of heavy ion radiation on GI epithelial cell migration up to two months after exposure and compare the results to gamma radiation. The generated data is expected to contribute towards developing GI risk estimates of heavy ion radiation, which is qualitatively high linear energy transfer (high-LET) in outer space relative to gamma-rays, which is low-LET. Mice (C57BL/6J) were exposed to sham, or 0.5 Gy of gamma or 56Fe radiation and small intestine, which is organized into well-defined crypts and villi and is considered a model system to study GI cell migration, and collected 7 and 60 d after exposure. Rate of IEC migration was assessed using BrdU pulse labeling 24 h prior to sacrificing the mice. We analyzed key signaling molecules involved in cell polarity, microtubule dynamics, and cell adhesion dynamics using immunoblots, immunofluorescence, immunohistochemistry. Our data from this study suggests that heavy ion radiation could impede GI epithelial cell migration persistently. Overall, these studies in normal GI tissues indicate a profound and persistent perturbation of normal GI physiology and cellular homeostasis, and has implications for space travel-related GI pathophysiological changes. At the genomics level we are using laboratory and bioinformatics tools of systems genomics to identify the features of gastrointestinal tumors arising in our murine models. Using next generation sequencing (NGS) based methods we are determining the global mutation burden, and mutation types including small mutations and structural variations in these tumors. To further inform mechanistic interpretation of HZE effects, we are planning to use NGS technologies to examine transcriptomic and epigenomic changes in colonocytes recovered after space radiation. Our overarching goal is to utilize integrative genomics analyses to identify the signatures of space radiation exposure, tumorigenesis, and the mechanistic consequences of these disturbances so that this information can be utilized to support risk assessment and mitigation studies. We are using formalin fixed paraffin embedded (FFPE) tumor samples for sequencing. As an initial step, FFPE sections were H&E stained for pathology evaluation and areas representing adenoma or carcinoma are marked for dissection. DNA is extracted from the dissected tissue using our standard protocols. We have developed a methodology for molecular quality control that will allow an estimate of sample purity and comprehensive sequencing for space radiation genomic signature identification. Analyses of mouse model data of simulated space radiation, using mechanistically-motivated mathematical models of carcinogenesis, are needed to better understand and potentially mitigate these risks. We have used data on heavy ions (12-C, 28-Si, and 56-Fe) and gamma-ray induced intestinal tumor frequency in APC1638N/+ mice. Our mathematical model includes: (1) targeted effects (TE), caused by ionizations within/close to nuclear DNA; (2) non-targeted effects (NTE), caused by radiation damage to other cell structures (e.g., mitochondria) and/or activation of stress signaling pathways; (3) cytotoxic effects. We used the model to estimate the NTE contribution to the dose response for each ion. According to our model, NTE cause the steep initial rise of the heavy ion dose responses observed between 0 and 0.1 Gy. Because at low fluences of heavy ions many cell nuclei will not be traversed by the cores of ion tracks, the contribution of NTE to the dose response at low doses is expected to be high. Model-based estimates of NTE contributions were consistent with this expectation: they dominated the dose response below 0.1 Gy. Non targeted effects may therefore play an important role in low-dose carcinogenesis from heavy ions. Overall, we plan to generate in vivo data on tumor frequency and mechanisms of tumor development using NASA mandated additional priority space radiation beams at lower doses relevant to space environment. These data will be used to reduce uncertainty in CRC risk prediction and develop reliable risk prediction model for human.
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